Optimization and Curve Sketching

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.02

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Critical points and the first/second derivative tests**

Find the critical points of $f(x) = x^3 - 6x^2 + 9x$, and classify each as a local max or min using the second derivative test.

$f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$, critical points $x = 1, 3$. $f''(x) = 6x - 12$: $f''(1) = -6 < 0$ (local max); $f''(3) = 6 > 0$ (local min).

2 **[Foundational]** • **Target: Critical points and the first/second derivative tests**

For $f(x) = x^4 - 4x^3$, find the intervals of concavity and any inflection points.

$f''(x) = 12x^2 - 24x = 12x(x-2)$. $f'' > 0$ (concave up) on $(-\infty, 0) \cup (2, \infty)$; $f'' < 0$ (concave down) on $(0, 2)$. Inflection points at $x = 0$ and $x = 2$.

3 **[Foundational]** • **Target: Optimization applications**

A farmer has 200 m of fencing to enclose a rectangular field against an existing wall (so only 3 sides need fencing). Find the dimensions that maximize the enclosed area.

Let x = width (two sides), y = length (one side): $2x + y = 200$, so $y = 200 - 2x$. Area $A = xy = x(200 - 2x) = 200x - 2x^2$. $A' = 200 - 4x = 0$ gives $x = 50$, $y = 100$. Maximum area 5000 m^2 at width 50 m, length 100 m.

4 **[Foundational]** • **Target: Optimization applications**

An open-top box is made from a 12 cm square sheet by cutting equal squares of side x from each corner and folding up the sides. Find the value of x that maximizes the box's volume.

$V(x) = x(12 - 2x)^2$, for $0 < x < 6$. $V'(x) = (12-2x)^2 + x \cdot 2(12-2x)(-2) = (12-2x)(12-6x)$. Setting $V' = 0$: $x = 6$ (rejected, gives $V = 0$) or $x = 2$. Maximum volume at $x = 2$ cm.

5 **[Foundational]** • **Target: More optimization**

A cylindrical can must have volume 500π cm^3. Using $V = \pi r^2 h$ and surface area $S = 2\pi r^2 + 2\pi rh$, find the radius that minimizes the material used (surface area).

From V: $h = \frac{500}{r^2}$. $S(r) = 2\pi r^2 + 2\pi r \cdot \frac{500}{r^2} = 2\pi r^2 + \frac{1000\pi}{r}$. $S'(r) = 4\pi r - \frac{1000\pi}{r^2} = 0$: $4r^3 = 1000$, $r^3 = 250$, $r = \sqrt[3]{250} \approx 6.30$ cm.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: More optimization

Find two positive numbers whose sum is 20 and whose product is as large as possible.

Let the numbers be x and $20 - x$: $P(x) = x(20 - x) = 20x - x^2$. $P'(x) = 20 - 2x = 0$ gives $x = 10$. Both numbers are 10, maximum product 100.

7 [Foundational] • Target: Curve sketching

For $f(x) = \frac{1}{3}x^3 - x^2 - 3x + 2$, find the intervals of increase and decrease.

$f'(x) = x^2 - 2x - 3 = (x - 3)(x + 1)$. Critical points $x = -1, 3$. Testing: increasing on $(-\infty, -1)$, decreasing on $(-1, 3)$, increasing on $(3, \infty)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Curve sketching

Using the results above, classify $x = -1$ and $x = 3$ as local maxima or minima via the first derivative test.

f changes from increasing to decreasing at $x = -1$: local maximum. f changes from decreasing to increasing at $x = 3$: local minimum.

9 [Intermediate] • Target: Minimizing distance and cost

Find the point on the parabola $y = x^2$ closest to the point $(0, 3)$, by minimizing $D^2 = x^2 + (x^2 - 3)^2$.

$D^2(x) = x^2 + (x^2 - 3)^2$. $\frac{d}{dx}D^2 = 2x + 2(x^2 - 3)(2x) = 2x[1 + 2(x^2 - 3)] = 2x(2x^2 - 5)$. Setting to 0: $x = 0$ or $x^2 = \frac{5}{2}$. Testing shows $x = 0$ gives a local max (distance 3); $x = \pm\sqrt{5/2}$ gives the minimum, at point $\left(\sqrt{5/2}, \frac{5}{2}\right)$.

10 [Intermediate] • Target: Minimizing distance and cost

A rectangular box with a square base and open top must have volume 32 m^3. Find the base side length that minimizes the surface area, using $S = x^2 + \frac{128}{x}$ (derived from $V = x^2 h = 32$).

$S'(x) = 2x - \frac{128}{x^2} = 0$: $2x^3 = 128$, $x^3 = 64$, $x = 4$ m.

11 [Intermediate] • Target: Sketching using derivative information

For $f(x) = 3x^4 - 4x^3$, find all critical points and classify each using the second derivative test.

$f'(x) = 12x^3 - 12x^2 = 12x^2(x - 1)$: critical points $x = 0, 1$. $f''(x) = 36x^2 - 24x$: $f''(0) = 0$ (test inconclusive, needs first derivative test — f' doesn't change sign at $x = 0$, so it's neither a max nor min); $f''(1) = 36 - 24 = 12 > 0$: local minimum at $x = 1$.

© ClipSAT — clipsat.org | All Rights Reserved

12 [Intermediate] • Target: Sketching using derivative information

For $f(x) = xe^{-x}$, find the critical point and determine whether it is a maximum or minimum.

$f'(x) = e^{-x} - xe^{-x} = e^{-x}(1-x) = 0$ gives $x = 1$. $f''(x) = -e^{-x}(1-x) - e^{-x} = e^{-x}(x-2)$; $f''(1) = e^{-1}(-1) < 0$: local maximum at $x = 1$.

13 [Intermediate] • Target: Visualizing critical points on a graph

The graph shows $f(x) = x^3 - 6x^2 + 9x$ from the first problem. Use the graph to confirm the local max at $x = 1$ and local min at $x = 3$ found algebraically.

The graph rises to a peak at $x = 1$ (local maximum, $f(1) = 4$), then falls to a valley at $x = 3$ (local minimum, $f(3) = 0$), matching the second-derivative-test classification.

14 [Intermediate] • Target: Optimization with a fixed perimeter

A rectangular garden must have a perimeter of 80 m. Find the dimensions that maximize its area.

$2x + 2y = 80$, so $y = 40 - x$. $A(x) = x(40-x) = 40x - x^2$. $A'(x) = 40 - 2x = 0$ gives $x = 20$, $y = 20$. The maximum-area rectangle is a 20×20 square, area 400 m^2.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Optimization with a fixed perimeter • Cross-Lesson: 16.01 Vector Fields and Line Integrals

A Norman window (a rectangle topped by a semicircle) has perimeter 10 m, with rectangle width x and height y: $2y + x + \pi\frac{x}{2} = 10$. Express the total area $A = xy + \frac{\pi}{8}x^2$ in terms of x alone, and state the equation you would set $A'(x) = 0$ to solve (do not solve).

From the constraint: $y = \frac{10 - x - \frac{\pi x}{2}}{2} = 5 - \frac{x}{2} - \frac{\pi x}{4}$. Substituting: $A(x) = x(5 - \frac{x}{2} - \frac{\pi x}{4}) + \frac{\pi}{8}x^2 = 5x - \frac{x^2}{2} - \frac{\pi x^2}{4} + \frac{\pi x^2}{8}$. Setting $A'(x) = 5 - x - \frac{\pi x}{2} + \frac{\pi x}{4} = 0$ would give the optimal x.

16 [Challenging] • Target: Concavity and the second derivative test in context • Cross-Lesson: 2.01 Evaluating Limits

A company's profit is modeled by $P(x) = -2x^2 + 80x - 300$, where x is units sold (hundreds). Find the production level that maximizes profit, and verify it is a maximum using the second derivative.

$P'(x) = -4x + 80 = 0$ gives $x = 20$. $P''(x) = -4 < 0$ for all x, confirming a maximum at $x = 20$ (2000 units), with maximum profit $P(20) = -800 + 1600 - 300 = 500$.

17 [Challenging] • Target: Curve sketching • Cross-Lesson: 2.02 Continuity

Local min at $x = c$ by First Derivative Test requires f':.

Final Answer: negative then positive. Applying Applications of Derivatives principles, Sign change from $-$ to $+$. Therefore, the verified correct option is negative then positive.

© ClipSAT — clipsat.org | All Rights Reserved

18 [Challenging] • Target: Minimizing distance and cost • Cross-Lesson: 3.01 The Derivative from First Principles

The Second Derivative Test says $f''(c) < 0$ at a critical point c implies:.

Final Answer: local maximum. Concave down ($f''(c) < 0$) at a critical point means a local maximum.

19 [Challenging] • Target: Sketching using derivative information • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

If $f'(x) = 0$ and $f''(x) = 0$, the Second Derivative Test is:.

Final Answer: inconclusive. Test fails; need higher derivatives or sign analysis.

20 [Challenging] • Target: Visualizing critical points on a graph • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Second Derivative Test: $f'(c) = 0$ and $f''(c) > 0 \Rightarrow x = c$ is a:.

Final Answer: local min. $f'' > 0$ at critical point $\Rightarrow$ local min.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Critical Points and First Derivative Test

Find all critical points of $f(x) = 2x^3 - 9x^2 + 12x - 5$.

a. $x = 1$ **and** $x = 2$ ✓

b. $x = -1$ and $x = -2$

c. $x = 3$ and $x = 0$

d. $x = 2$ only

$f'(x) = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2) = 6(x - 1)(x - 2) = 0 \Rightarrow x = 1$ and $x = 2$.

22 [Foundational] • Target: Critical Points and First Derivative Test

Classify the local extrema of $f(x) = 2x^3 - 9x^2 + 12x - 5$.

a. **Local max at $x = 1$; local min at $x = 2$** ✓

b. Local min at $x = 1$; local max at $x = 2$

c. Inflection point at $x = 1$

d. Local max at $x = 2$ only

$f'(x)$ is positive for $x < 1$, negative for $1 < x < 2$, and positive for $x > 2$. By the First Derivative Test, $x = 1$ is a local max and $x = 2$ is a local min.

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Critical Points and First Derivative Test

Find the absolute maximum of $g(x) = 4x - x^2$ on the closed interval $[0, 3]$.

a. 4 ✓

b. 3

c. 0

d. 5

$g'(x) = 4 - 2x = 0 \Rightarrow x = 2$. Test critical point and endpoints: $g(0) = 0, g(2) = 4, g(3) = 3$. The absolute maximum is 4 at $x = 2$.

24 [Foundational] • Target: Concavity and the Second Derivative Test

Determine the interval(s) where $f(x) = x^4 - 4x^3 + 10$ is concave upward.

a. $(-\infty, 0) \cup (2, \infty)$ ✓

b. $(0, 2)$

c. $(2, \infty)$

d. $(-\infty, 2)$

$f'(x) = 4x^3 - 12x^2$. $f''(x) = 12x^2 - 24x = 12x(x - 2) > 0 \Rightarrow x < 0$ or $x > 2$.

25 [Foundational] • Target: Concavity and the Second Derivative Test

Find the x-coordinates of all inflection points of $y = x^3 - 6x^2 + 9x$.

a. $x = 2$ ✓

b. $x = 1$ and $x = 3$

c. $x = 0$

d. $x = 4$

$y' = 3x^2 - 12x + 9$. $y'' = 6x - 12 = 0 \Rightarrow x = 2$. Concavity changes from negative to positive at $x = 2$.

26 [Foundational] • Target: Concavity and the Second Derivative Test

If $f'(3) = 0$ and $f''(3) = -5$, what does the Second Derivative Test conclude about $x = 3$?

a. f **has a local maximum at** $x = 3$ ✓

b. f has a local minimum at $x = 3$

c. $x = 3$ is an inflection point

d. The test is inconclusive

Since $f'(3) = 0$ and $f''(3) < 0$, the graph is concave down at the critical point, indicating a local maximum.

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Applied Optimization: Perimeter, Area, and Volume

A farmer has 120 m of fencing to enclose a rectangular field adjacent to a river (no fence needed along the river). What dimensions maximize the enclosed area?

a. **Width** $= 30$ m**, Length** $= 60$ m ✓

b. Width $= 40$ m, Length $= 40$ m

c. Width $= 20$ m, Length $= 80$ m

d. Width $= 25$ m, Length $= 70$ m

Perimeter: $2w + l = 120 \Rightarrow l = 120 - 2w$. Area: $A(w) = w(120 - 2w) = 120w - 2w^2$.
$A'(w) = 120 - 4w = 0 \Rightarrow w = 30$ m, $l = 60$ m.

28 [Foundational] • Target: Applied Optimization: Perimeter, Area, and Volume

An open-top rectangular box is made from a 16 by 10 inch cardboard sheet by cutting equal squares of side x from each corner and folding up the sides. What x maximizes the box volume?

a. 2 inches ✓

b. 3 inches

c. 1.5 inches

d. 2.5 inches

$V(x) = x(16 - 2x)(10 - 2x) = 4x^3 - 52x^2 + 160x$.
$V'(x) = 12x^2 - 104x + 160 = 4(3x^2 - 26x + 40) = 4(3x - 20)(x - 2) = 0$. Since $x < 5$, $x = 2$ inches.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Applied Optimization: Perimeter, Area, and Volume

A closed cylindrical can must hold 1000 cm^3 of liquid. What relationship between height h and radius r minimizes total surface area?

a. $h = 2r$ ✓

b. $h = r$

c. $h = 4r$

d. $h = \pi r$

$S = 2\pi r^2 + 2\pi rh$. With $V = \pi r^2 h = 1000 \Rightarrow h = 1000/(\pi r^2)$. $S(r) = 2\pi r^2 + 2000/r$.
$S'(r) = 4\pi r - 2000/r^2 = 0 \Rightarrow 4\pi r^3 = 2000 = 2V = 2\pi r^2 h \Rightarrow h = 2r$.

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Economics and Distance Optimization

A company's revenue and cost functions are $R(x) = 50x - 0.5x^2$ and $C(x) = 10x + 200$. What production level x maximizes profit?

a. 40 ✓

b. 50

c. 30

d. 60

Profit $P(x) = R(x) - C(x) = -0.5x^2 + 40x - 200$. $P'(x) = -x + 40 = 0 \Rightarrow x = 40$.

31 [Intermediate] • Target: Economics and Distance Optimization

Find the point on the line $y = 2x + 1$ that is closest to the point $(4, 4)$.

a. $(2, 5)$ ✓

b. $(1, 3)$

c. $(3, 7)$

d. $(1.5, 4)$

Squared distance $D(x) = (x-4)^2 + (2x+1-4)^2 = (x-4)^2 + (2x-3)^2 = x^2 - 8x + 16 + 4x^2 - 12x + 9 = 5x^2 - 20x + 25$. $D'(x) = 10x - 20 = 0 \Rightarrow x = 2, y = 5$.

32 [Intermediate] • Target: Economics and Distance Optimization

A Norman window has the shape of a rectangle surmounted by a semicircle. If the perimeter is 16 ft, what radius r of the semicircle maximizes the light admitted?

a. $\frac{16}{4+\pi}$ ft ✓

b. $\frac{8}{2+\pi}$ ft

c. 2 ft

d. $\frac{16}{\pi}$ ft

Perimeter: $2r + 2h + \pi r = 16 \Rightarrow 2h = 16 - (2+\pi)r$. Area: $A = 2rh + \frac{1}{2}\pi r^2 = r(16 - (2+\pi)r) + \frac{1}{2}\pi r^2 = 16r - (2 + \pi/2)r^2$. $A'(r) = 16 - (4+\pi)r = 0 \Rightarrow r = \frac{16}{4+\pi}$ ft.

33 [Intermediate] • Target: Economics and Distance Optimization

The second derivative test fails ($D = 0$). This means the critical point is:

a. always a saddle

b. **indeterminate without further analysis** ✓

c. always min

d. always max

$D = 0$: inconclusive — need higher-order analysis.

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Critical points and the first/second derivative tests

Optimal rectangle of fixed perimeter P has max area when:

a. length $= 2 \times$ width

b. **it is a square ✓**

c. width $= P$

d. length $= P/4$

Square maximises area for fixed perimeter.

35 [Intermediate] • Target: Optimization applications

Consider the region bounded by the curve $y = x^2$, the x-axis, and the vertical lines $x = 1$ and $x = 3$. What is the area of this region?

a. $\frac{26}{3}$ ✓

b. $\frac{28}{3}$

c. $\frac{25}{3}$

d. $\frac{27}{3}$

The area of a region bounded by a curve $y = f(x)$, the x-axis, and the vertical lines $x = a$ and $x = b$ is given by the definite integral $\int_a^b f(x)\,dx$. In this case, $f(x) = x^2$, $a = 1$, and $b = 3$. The area A is given by: $A = \int_1^3 x^2\,dx$ To evaluate this integral, we find the antiderivative of x^2, which is $\frac{x^3}{3}$. Now, we apply the Fundamental Theorem of Calculus.

36 [Intermediate] • Target: More optimization

$f(x, y) = xy$: at origin, the second derivative test gives $D = f_{xx}f_{yy} - f_{xy}^2 = 0 - 1 = -1 < 0 \Rightarrow$

a. **saddle point ✓**

b. local max

c. local min

d. inconclusive

Applying Partial Derivatives and Gradient principles, $D < 0 \Rightarrow$ saddle. Therefore, the verified correct option is saddle point.

37 [Intermediate] • Target: Curve sketching

Signed area and winding number: the winding number of curve C around origin is $\frac{1}{2\pi}\oint_C \frac{x\,dy - y\,dx}{x^2 + y^2} =$

a. always 1

b. **an integer ✓**

c. always 0

d. always π

Applying Applications of Integration principles, Winding number is an integer. Therefore, the verified correct option is an integer.

© ClipSAT — clipsat.org | All Rights Reserved

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Minimizing distance and cost • Cross-Lesson: 3.01 The Derivative from First Principles

Second derivative of $f(x) = x^4$:

a. $4x^3$

b. $12x^2$ ✓

c. $12x$

d. $24x$

Applying Applications of Derivatives principles, $f' = 4x^3$, $f'' = 12x^2$. Therefore, the verified correct option is $12x^2$.

39 [Challenging] • Target: Sketching using derivative information • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Moments of inertia: second moment of area $I_x = \int y^2\,dA$ for rectangle $b \times h$ about its centroidal axis:

a. $bh^3/12$ ✓

b. $bh^3/3$

c. $bh^2/12$

d. $b^2h/12$

Applying Applications of Integration principles, $I_x = bh^3/12$. Therefore, the verified correct option is $bh^3/12$.

40 [Challenging] • Target: Visualizing critical points on a graph • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

For polar curve $r = e^\theta$, $\theta \in [0, 2\pi]$, the area swept is:

a. $e^{4\pi}$

b. $\frac{1}{2}(e^{4\pi} - 1)$

c. $\frac{1}{4}(e^{4\pi} - 1)$ ✓

d. $\pi e^{4\pi}$

$A = \frac{1}{2}\int_0^{2\pi} e^{2\theta} d\theta = \frac{1}{4}[e^{2\theta}]_0^{2\pi} = \frac{e^{4\pi}-1}{4}$.

41 [Challenging] • Target: Optimization with a fixed perimeter • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

If $f'(x) = 0$ and $f''(x) = 0$ at $x = c$, the Second Derivative Test is:

a. guaranteed a maximum

b. guaranteed an inflection point

c. guaranteed a minimum

d. **inconclusive** ✓

When $f''(c) = 0$ too, the test gives no information; another method is needed.

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Concavity and the second derivative test in context • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Find all critical points of $f(x) = x^{2/3}(x - 2)$:

a. $x = 0$ only

b. $x = 2$ only

c. $x = 0$ **and** $x = 4/5$ ✓

d. $x = 0$ and $x = 2$

$f' = \frac{5x-4}{3x^{1/3}} = 0 \Rightarrow x = 4/5$; also undefined at $x = 0$.

43 [Challenging] • Target: Critical Points and First Derivative Test • Cross-Lesson: 6.01 Related Rates

Fence three sides of rectangle next to wall (wall is 4th side). Area $= xy = A$, perimeter $= x + 2y = L$. Max area:

a. $L^2/8$ ✓

b. $L^2/4$

c. $L^2/6$

d. $L^2/2$

$x = L - 2y$; $A = (L - 2y)y$; max at $y = L/4$; $A_{max} = L^2/8$.

44 [Challenging] • Target: Concavity and the Second Derivative Test • Cross-Lesson: 7.01 Integration by Substitution

Critical points occur where $f'(x)$ is:

a. positive

b. negative

c. **zero or undefined** ✓

d. equal to $f(x)$

Critical points require $f' = 0$ or f' undefined.

45 [Challenging] • Target: Applied Optimization: Perimeter, Area, and Volume • Cross-Lesson: 7.02 Integration by Parts

Second derivative test: $f''(c) > 0$ implies $x = c$ is:

a. **local min** ✓

b. local max

c. inflection

d. saddle

$f'' > 0$: concave up $\Rightarrow$ local min.

© ClipSAT — clipsat.org | All Rights Reserved